

1. Administrative & Study Identification

Full Study Title: A Phase III, Open-Label, Multi-Center, Randomized Study Comparing AAA817+ARPI Versus Standard of Care in Adult Participants with PSMA-Positive Metastatic Castration Resistant Prostate Cancer **Short Title/Acronym:** AcTFirst **Protocol Number:** CAAA817B12301 **NCT Number:** NCT06855277 **Sponsor Name:** Novartis **Investigational Product:** [225Ac]Ac-PSMA-617 (AAA817) and Androgen Receptor Pathway Inhibitor (ARPI) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine whether [225Ac]Ac-PSMA-617 (AAA817) plus androgen receptor pathway inhibitor (ARPI) improves radiographic progression-free survival (rPFS) compared to the investigator's choice of standard of care (SOC) in adult participants with PSMA-positive mCRPC. **Secondary Objectives:** To evaluate overall survival (OS), additional progression-free survival metrics, safety, and tolerability of the investigational combination therapy.

2.2 Background & Rationale To address the therapeutic needs of patients with PSMA-positive metastatic castration-resistant prostate cancer (mCRPC) who have documented progressive disease while on prior ARPI treatment but have not yet been exposed to taxane-containing chemotherapy in the mCRPC setting, nor any prior PSMA-targeting radioligand therapy. The study aims to establish the superiority of AAA817+ARPI over changing ARPI, chemotherapy, or [177Lu]Lu-PSMA-617.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard of Care: ARPI change, taxane-based chemotherapy, or [177Lu]Lu-PSMA-617) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Trial Status: Recruiting **Target \$N\$:** 605 **Number of Sites:** Approximately 55 locations globally **Estimated Study Start:** 01-Jul-2025 **Estimated Primary Completion:** 29-Sep-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age with an ECOG performance status of 0 to 2. 2. Histological and/or cytological confirmation of adenocarcinoma of the prostate with PSMA-PET positive disease using an approved PSMA imaging agent. 3. Diagnosed with mCRPC with documented progressive disease while on treatment with ARPI in mHSPC or earlier setting as their last treatment.

4.2 Exclusion Criteria 1. Participants with mixed histology (neuroendocrine). 2. Previous taxane-based chemotherapy in the mCRPC setting (allowed in mHSPC setting). 3. Previous anti-cancer treatment with any approved or investigational radiopharmaceuticals (e.g., [177Lu]Lu-PSMA, [177Lu]-DOTA, or Radium-223), or external beam radiotherapy within 6 weeks of randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AAA817 given for up to 6 cycles at a dose of 10 Megabecquerel (MBq) $\pm$ 10% combined with an oral ARPI (enzalutamide or abiraterone) per investigator's choice. **Route of Administration:** Intravenous (AAA817) and Oral (ARPI). **Duration of Treatment:** Up to 6 cycles for AAA817; ARPI to continue as per protocol end-of-treatment criteria.

5.2 Concomitant & Prohibited Therapy Permitted: Supportive care at the discretion of the investigator according to best institutional practice for mCRPC, including ongoing androgen deprivation therapy (ADT). **Prohibited:** Any other approved or investigational systemic therapy (chemotherapy, immunotherapy, biologics) within 28 days or 5 half-lives before randomization. Crossover is not allowed among study arms.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Eligibility Assessment, PSMA-PET/CT Scan
Baseline	Day 0	Randomization	Interim
Cycles 1-6	AAA817 IV Administration, ARPI Treatment, Routine Safety Assessments	End of Study	Follow-up Final Safety Evaluation, Follow-up for rPFS/survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Radiographic Progression Free Survival (rPFS): Time to radiographic disease progression or

death due to any cause, evaluated to demonstrate the superiority of the AAA817+ARPI arm. **Measurement Method:** Imaging assessment (e.g., CT/MRI and bone scans) centrally or locally reviewed.

7.2 Secondary & Exploratory Endpoints Key Secondary Endpoints:

Overall Survival (OS), Radiographic Progression Free Survival by PSMA PET/CT (rPFS-PET), and Progression Free Survival (PFS).

Other Assessments: General safety, tolerability, and clinical/PSA response.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Safety will be assessed routinely during the study via onsite visits and laboratory workups.

Serious Adverse Events (SAEs): Reported per standard regulatory guidelines (typically within 24 hours). **Data Monitoring Committee (DMC):** Supervised by Novartis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy and Safety Set for AEs. **Sample Size Justification:** Powered to detect a statistically significant prolongation in rPFS for the AAA817+ARPI group over the SOC group, targeting an enrollment of 605 participants. **Handling of Missing Data:** Standard handling (e.g., censoring for rPFS) per protocol guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine clinical monitoring at participating sites globally. **Audit Trail:** eCRF entry tracking, query resolution, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CAAA817B12301 **Registry Number:** NCT06855277 **Posting ID:** 868829634600770456 **Sponsor/Lead Organization:** Novartis **Study Title:** ActFirst **Official Regulatory Title:** A Phase III, Open-Label, Multi-Center, Randomized Study Comparing AAA817+ARPI Versus Standard of Care in Adult Participants with PSMA-Positive Metastatic Castration Resistant Prostate Cancer

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Disease Area:** Prostate Cancer **Diagnosis Threshold:** PSMA-PET positive disease **Medical Methodology:** Targeted Radioligand Therapy + Androgen Receptor Pathway Inhibitor **Optimization Target:** Prolongation of Radiographic progression-free survival (rPFS)

13. Study Procedures & Methodology

Management Pathway: Targeted Radioligand Therapy vs. Standard of Care **Education Protocol (HCP):** Site initiation and radioactive material handling/protocol training **Education Protocol (Patient):** Informed consent and radiation safety instructions post-treatment **Quality Audit Mechanism:** Clinical monitoring and strict adherence tracking to protocol end-of-treatment criteria

14. Observation & Follow-Up Schedule

Total Duration: Approximately 39 months (From July 2025 to September 2028) **Onsite Visit Cadence:** Every cycle for up to 6 cycles, followed by standard surveillance **Remote Monitoring Frequency:** As required by protocol for survival/rPFS tracking **Checklist Review Interval:** Routine intervals per protocol

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				